IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

Received 6 September 2025, accepted 26 September 2025, date of publication 30 September 2025,
date of current version 17 October 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3616056

RESEARCH ARTICLE

Encouraging Discriminative Attention Through Contrastive Explainability Learning for Lung Cancer Diagnosis

V. SHRAVYA[1], MEGHANA SUNIL[1], B. NATARAJAN[1], AND R. ELAKKIYA[2], (Member, IEEE)
[1]Vellore Institute of Technology, Chennai 600127, India
[2]Birla Institute of Technology and Science Pilani, Dubai, United Arab Emirates

Corresponding author: B. Natarajan (natarajan.b@vit.ac.in)

ABSTRACT Lung cancer diagnosis using CT scans is critical for early detection, but existing deep methods often lack interpretability, especially in highlighting medically relevant regions. Most approaches optimize only for prediction accuracy, leaving model explanations unstructured and inconsistent across samples. We introduce Contrastive Explainability Learning (CEL), a novel training approach that aligns Grad-CAM heatmaps across class-consistent samples while enforcing dissimilarity across different classes. Unlike prior methods, CEL integrates explanation supervision directly into the loss function, enabling interpretable representation learning without sacrificing accuracy. Using only a lightweight, spatially-attended CNN, our model achieves strong performance (99.2% accuracy, 99.5% F1 score) on the IQ-OTH/NCCD dataset and demonstrates robust generalization (93.0% accuracy) on the more complex HF Lung Cancer dataset with multiple cancer subtypes. Statistical analysis across multiple trials confirms that these improvements are significant ($p < 0.01$). We demonstrate through comprehensive comparisons with alternative XAI methods that CEL produces more consistent, discriminative explanations with minimal computational overhead. Experiments show that our contrastive saliency framework guides the CNN to focus on class-specific anatomical regions, improving both transparency and diagnostic trust while maintaining efficiency suitable for clinical deployment.

INDEX TERMS Lung cancer diagnosis, explainable AI, contrastive learning, saliency maps, medical image analysis.

I. INTRODUCTION

Lung cancer remains one of the leading causes of cancer-related mortality worldwide, with survival outcomes heavily dependent on early and accurate diagnosis. Traditionally, diagnostic assessment has relied on handcrafted features derived from medical images, histopathological slides, or genomic data [1], [2]. Statistical models such as Cox proportional hazards regression and survival analysis techniques have been commonly employed to predict patient outcomes. While these methods have demonstrated reasonable performance, they heavily depend on expert-driven feature selection, which can be biased and may fail to capture subtle yet crucial variations in imaging or molecular data [3], [4].

Deep learning has revolutionized oncology by automating feature extraction across various diagnostic modalities, including computed tomography (CT), chest X-ray, magnetic resonance imaging (MRI), positron emission tomography (PET), histopathology, and genomic profiling [1]. In lung cancer diagnosis, convolutional neural networks (CNNs) and transformer-based architectures have shown great promise in detecting, classifying, and subtyping tumors with high accuracy. Recent advances have leveraged hybrid models combining CNNs and Vision Transformers (ViTs) with sophisticated attention mechanisms to capture both fine-grained local features and global contextual information in CT scans [2]. Such models have demonstrated state-of-the-art performance

The associate editor coordinating the review of this manuscript and approving it for publication was Abdullah Iliyasu.

© 2025 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License.
For more information, see https://creativecommons.org/licenses/by/4.0/

in differentiating benign and malignant nodules, as well as identifying specific cancer subtypes, while maintaining lightweight architectures suitable for clinical application. Ozdemir et al. [5] further advanced this approach through an attention-enhanced InceptionNeXt-based hybrid model achieving 99.54% accuracy on the IQ-OTH/NCCD dataset by optimizing grid and block attention mechanisms for multi-scale feature processing. Similar hybrid architectures have shown promise beyond lung cancer, with Pacal and Attallah developing InceptionNeXt-Transformer models for breast cancer diagnosis [6] and colorectal cancer detection [7], highlighting a growing trend toward architectures that effectively balance local feature extraction with global context modeling.

Early detection of lung nodules using CNN-based models has shown promising results, with architectures like U-Net and two-stage CNNs effectively segmenting and classifying nodules in CT scans [8]. Histopathological image analysis has also benefited from CNNs, with studies demonstrating high accuracy in classifying lung cancer subtypes, such as adenocarcinoma and squamous cell carcinoma [4]. Additionally, multimodal approaches integrating imaging, clinical, and genomic data have been explored to enhance predictive accuracy [9]. However, these methods primarily rely on correlational patterns rather than learning clinically relevant tumor characteristics, leading to predictions misaligned with medical intuition [3], [4]. While CNN-based architectures have achieved high classification accuracy, they struggle with interpretability, making it difficult for clinicians to understand the reasoning behind diagnostic assessments [10], [11]. A naive solution involves post hoc explanation methods, such as saliency maps, but these primarily highlight spatial features without explicitly identifying which regions are most critical for lung cancer diagnosis or understanding why those regions are important, limiting their clinical reliability [12].

Despite these successes, deep learning models often struggle with interpretability, as their decision-making processes remain largely opaque to clinicians [10], [11]. This lack of transparency limits trust and hinders adoption in medical practice. Post hoc explanation methods, such as saliency maps and Grad-CAM, provide some insight by highlighting important spatial features, but they frequently suffer from instability and fail to explicitly identify medically relevant regions critical for diagnosis [12]. Moreover, attention mechanisms from transformer-based architectures, as explored in other domains, highlight the potential to capture both local and global dependencies in a computationally efficient manner while improving interpretability.

To address these challenges, we introduce Contrastive Explainability Learning (CEL), a novel training framework that integrates explanation supervision directly into the learning process. CEL enforces contrastive constraints on Grad-CAM heatmaps by encouraging consistency of attention maps across samples within the same class, while promoting dissimilarity across different classes. This approach guides the model to focus on class-specific anatomical regions, producing interpretable and clinically aligned saliency maps without sacrificing classification accuracy. As illustrated in Figure 1, CEL leads to stable, transparent predictions, enhancing diagnostic trust and making AI-driven lung cancer detection more reliable

To summarize, our contributions are as follows:

- We introduce a novel training strategy that enforces consistency of Grad-CAM heatmaps across same-class images while encouraging divergence across different classes, establishing a structured explanation embedding space with clear class boundaries that grounds model attention in class-specific anatomical semantics.
- We propose a hybrid loss function with a margin-based contrastive explainability term that specifically targets the saliency map distribution rather than feature representations, enabling direct optimization of explanation quality during training—a significant departure from post-hoc explanation methods that addresses the fundamental limitations of current XAI approaches in medical imaging.
- Our approach integrates explanation supervision directly into the optimization objective without requiring architectural modifications, allowing attention maps to evolve naturally alongside feature learning while maintaining computational efficiency (only 18% training overhead with no inference penalty), making it particularly suitable for clinical deployment where both interpretability and performance are critical.
- We demonstrate through comprehensive comparative analysis with 16 vision transformer architectures, 19 CNN variants, and multiple XAI techniques that our approach provides statistically significant improvements in both diagnostic accuracy and explanation quality, verified through rigorous significance testing ($p < 0.01$) across multiple independent trials.

The remainder of this paper is organized as follows: Section II reviews related work in medical image classification, attention mechanisms, and explainable AI approaches. Section III details our proposed method, including the attention-guided feature encoding framework, the contrastive explainability learning formulation, and the composite loss function. Section IV presents our experimental setup, covering datasets, implementation details, and reports comprehensive results. Finally, Section V concludes with a summary of our contributions and discusses future research directions.

II. RELATED WORKS

A. CONTRASTIVE LEARNING IN MEDICAL IMAGING

Contrastive self-supervised learning (SSL) has emerged as a pivotal technique in medical image analysis, addressing challenges associated with limited annotated data. Wang et al. [13] presented a review on contrastive SSL techniques originally developed for natural images and their adaptation to medical imaging, emphasizing their potential in surpassing supervised learning in data-scarce environments.

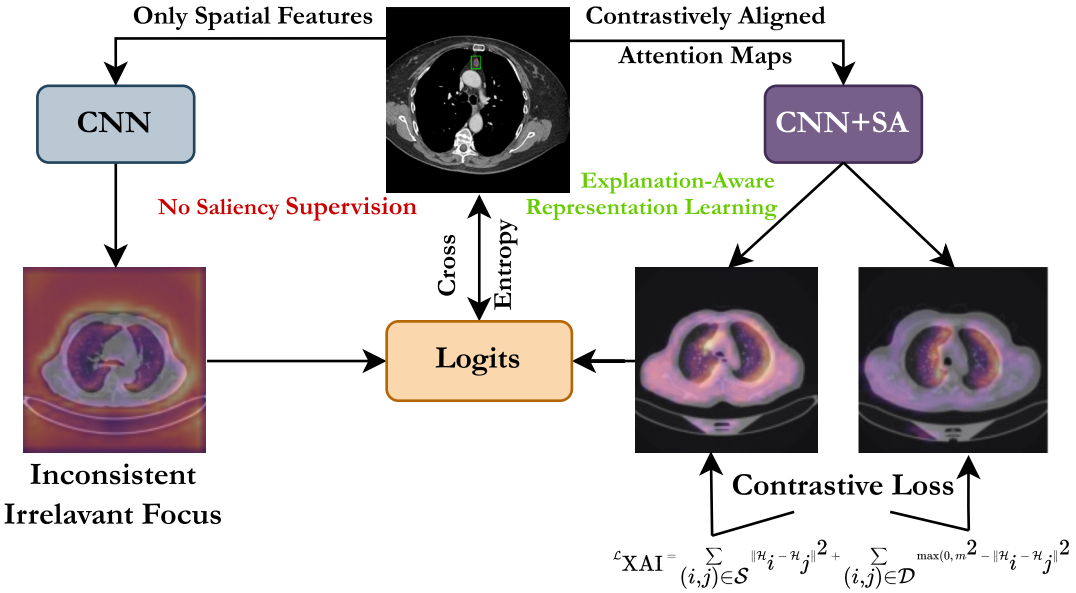


FIGURE 1. Overview of our proposed CEL framework. Unlike standard CNNs trained solely for prediction, our model incorporates a spatial attention module and enforces saliency alignment across samples. Grad-CAMs from the baseline model (left) are inconsistent and diffuse, whereas our CEL-guided architecture (right) yields class-consistent and diagnostically relevant focus maps. The combined objective supervises both predictions and explanations using a composite loss comprising cross-entropy and contrastive explainability loss $\mathcal{L}_{\text{XAI}}$.

Zeng et al. [14] tackled segmentation challenges by introducing Positional Contrastive Learning (PCL), leveraging spatial context to mitigate false negative pairs, thereby enhancing segmentation accuracy. Hu et al. [15] extended this approach by incorporating semi-supervised contrastive learning with limited labels, proposing a supervised local contrastive loss to improve feature representation while mitigating computational constraints through downsampling and block division. Xiang et al. [16] refined semi-supervised segmentation further by introducing self-ensembling contrastive learning, implementing teacher-student networks and anatomical-aware sampling strategies to optimize feature discrimination.

Federated learning (FL) combined with contrastive learning has shown promise in distributed medical image segmentation. Wu et al. [17] proposed a Federated Contrastive Learning (FCL) framework, overcoming data diversity limitations across sites by exchanging features instead of raw data, aligning local and global features through structural matching. Lu et al. [18] explored the synergy between transfer learning and contrastive learning, demonstrating that integrating BiT with SimSiam accelerates convergence while maintaining superior performance. Liu et al. [19] introduced CLMorph, a contrastive registration-based model for unsupervised segmentation, effectively bridging the gap between supervised and unsupervised techniques by incorporating registration-driven segmentation. Hou et al. [20] developed the MKCL model for radiology report generation, integrating medical knowledge graphs with contrastive learning to enhance abnormality detection and mitigate textual bias.

B. CONVOLUTIONAL NEURAL NETWORKS FOR LUNG CANCER DIAGNOSIS

Deep learning techniques have significantly advanced lung cancer diagnosis through convolutional neural networks (CNNs). Serj et al. [3] proposed a deep CNN model focusing on high-level image representation, achieving strong classification performance with minimal variance. Mangal et al. [4] extended this approach to histopathological images, developing a CNN-based system that differentiated lung and colon cancers with over 97% accuracy. Cao et al. [8] tackled the challenge of lung nodule detection by introducing a two-stage CNN architecture, leveraging UNet-based segmentation and dual-pooling classification networks, which enhanced both recall and precision. To?açar et al. [21] further improved classification accuracy by integrating CNNs with a feature selection method (mRMR). Unlike previous works, Hatt et al. [12] demonstrated that CNN-based classification of COPD and emphysema could be predictive of lung cancer, emphasizing the potential of comorbid condition analysis in diagnostic processes.

Recent studies have also focused on enhancing interpretability and multimodal learning in lung cancer diagnosis. Tomassini et al. [11] conducted a comprehensive survey on CNN applications, identifying key challenges such as data accessibility and reproducibility. Shen et al. [10] addressed the interpretability issue by developing a hierarchical semantic CNN, which incorporated radiological features into the network's decision-making process, making predictions more explainable for clinicians. Sangeetha [9] et al. introduced a multimodal fusion deep neural network (MFDNN) that

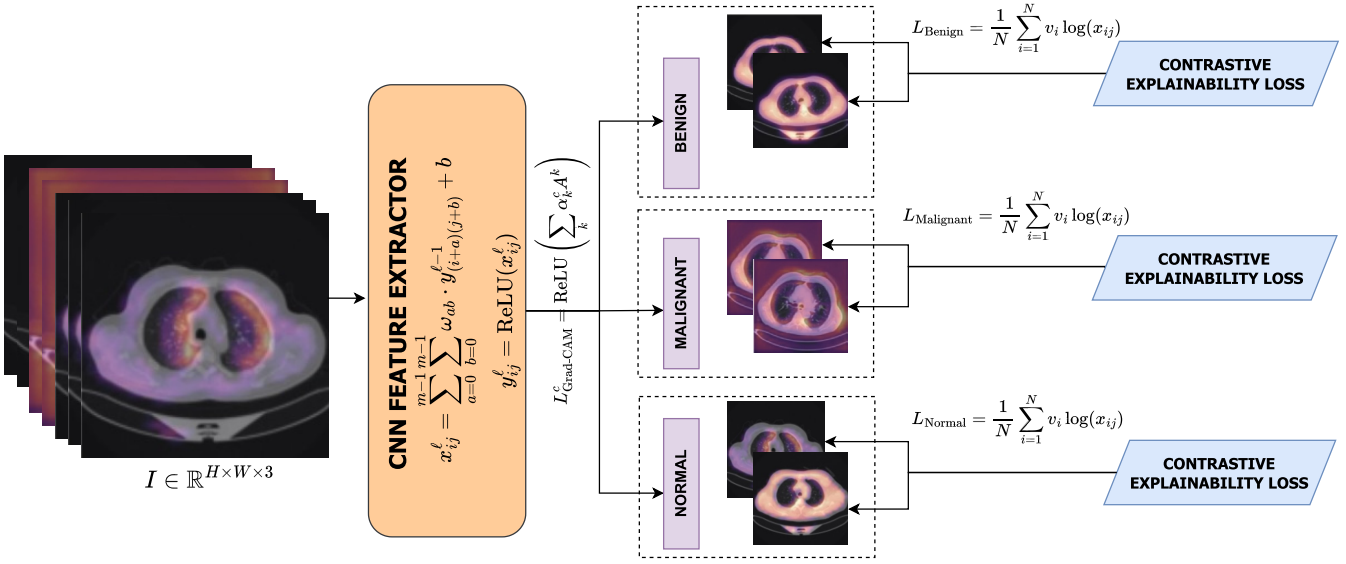


FIGURE 2. Pipeline of our CEL framework showing the integration of Grad-CAM constraints during CNN training. The system simultaneously optimizes classification accuracy and explanation consistency through the hybrid loss $\mathcal{L}_{\text{total}}$.

integrated imaging, genomic, and clinical data, achieving an accuracy of 92.5%, outperforming standard CNN and DNN models. These works highlight the ongoing shift from black-box models to more transparent and integrated AI-driven diagnostic tools.

C. EXPLAINABLE AI IN MEDICAL IMAGING

The growing need for transparency in AI-driven medical image analysis has led to significant advancements in explainable AI (XAI). Van der Velden et al. [22] introduced a framework to categorize XAI techniques in deep learning-based medical imaging, emphasizing the need for explainability in high-stakes decision-making. Borys et al. [23], [24] further refined this perspective by differentiating between saliency-based and non-saliency-based XAI methods. While the former focuses on feature attribution via visual heatmaps, the latter explores case-based, textual, and auxiliary explanations to improve clinical interpretability. Jin et al. [25] tackled multi-modal medical image analysis, applying both gradient- and perturbation-based post-hoc explanation techniques, but their reliance on post-hoc methods raises concerns about fidelity to model decisions. Hou et al. [26] proposed an alternative with self-explainable AI (S-XAI), embedding interpretability directly into deep learning models, enhancing transparency while mitigating concerns of post-hoc explanation validity.

Expanding beyond individual techniques, Muhammad and Bendechache [27] systematically reviewed state-of-the-art XAI methods, identifying key challenges in their adoption for clinical decision support. Band et al. [28] provided a comparative assessment of interpretability techniques, detailing various XAI methods like LIME, SHAP, and Grad-CAM, while discussing their applicability to different medical conditions. Holzinger et al. [29] approached XAI from a regulatory and usability standpoint, stressing the importance of explainability in the context of GDPR compliance and clinical trust.

To provide a comprehensive overview of the literature discussed above, Table 1 summarizes key related works, their problem statements, contributions, and limitations. This reference highlights how our work addresses existing gaps by integrating contrastive learning principles with explanation supervision in a unified framework.

D. EXPLANATION CONSISTENCY IN MEDICAL IMAGING

Expanding on recent trends in explainable AI (XAI), researchers have explored ways to make model explanations more consistent and trustworthy. ATCON [30] regularizes vision models by enforcing attention map similarity between related inputs, improving both classification and explanation reliability when paired with techniques like Grad-CAM and Integrated Gradients. Attention Consistency (AC) [33] extends this idea to a semi-supervised setting, encouraging agreement among attention distributions for similar samples to boost accuracy and efficiency. Liu et al. [34] embed human attention knowledge into saliency-based XAI for more plausible and clinically useful interpretations, while SHIELD [35] improves explanation quality by selectively masking inputs. Together, these methods advance stability and interpretability, especially in clinical imaging.

Our Contrastive Explainability Learning (CEL) framework differs fundamentally from these approaches in several key aspects. First, unlike ATCON and AC that focus primarily on local consistency, CEL establishes a dual-objective

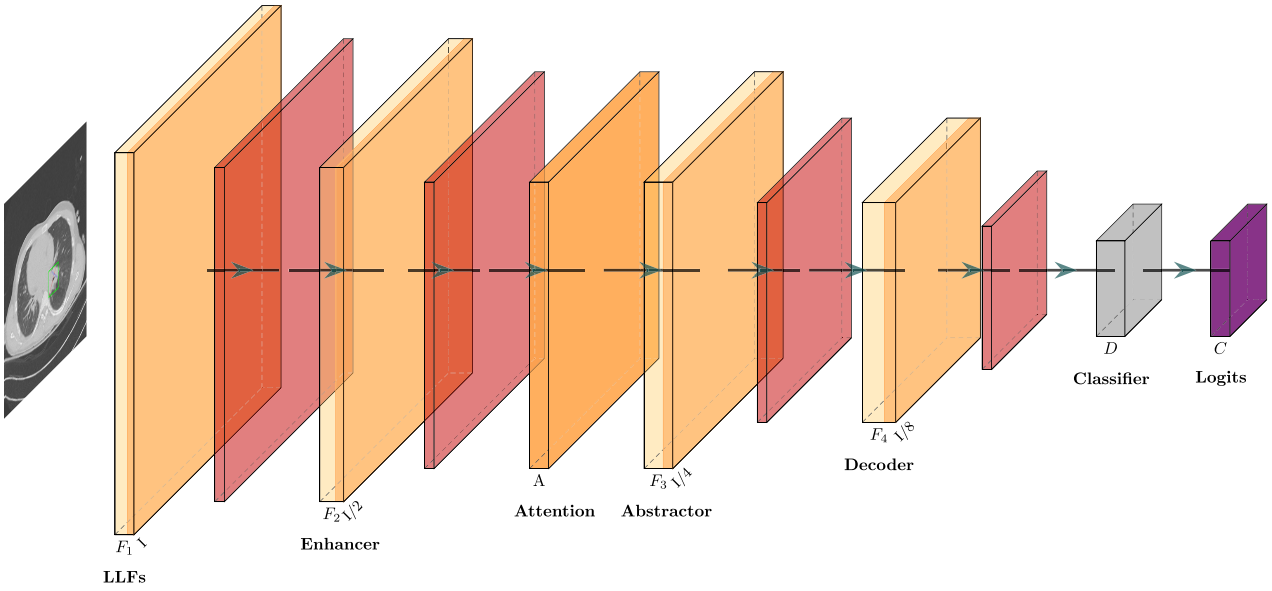


FIGURE 3. Overall architecture of our attention-guided CNN with contrastive explainability supervision. Low-level features (F_1) are progressively refined through convolutional encoding and attention-based enhancement. The spatial attention module generates an intermediate attention map A used to compute the attended representation F', which is further processed by a decoder and classifier. Grad-CAM explanations are extracted from the mid-level features for contrastive alignment.

mechanism that simultaneously enforces intra-class explanation alignment while maximizing inter-class explanation divergence. Second, while prior methods like [36] use contrastive explanations as post-hoc justifications for predictions, our approach integrates the contrastive objective directly into the training loss, allowing the model to learn discriminative explanations during feature extraction rather than as an afterthought. Third, our framework uniquely employs a margin-based contrastive mechanism specifically designed for explanation embeddings rather than feature embeddings, creating a structured explanation space with clear class-specific clusters. This distinction is critical, as it shifts the paradigm from treating explanations as byproducts of trained models to treating them as first-class learning objectives that actively shape model attention. Drawing inspiration from contrastive learning [31] and visualization techniques like Grad-CAM [32], CEL represents a novel synthesis that specifically addresses the unique challenges of medical image classification where explanation consistency and discriminative power are paramount for clinical trust and adoption.

III. PROPOSED METHOD

A. ATTENTION-GUIDED FEATURE ENCODING

Our design philosophy is guided by the premise that explainability supervision, when directly embedded into the training objective, can serve as a strong inductive signal - thereby reducing the need for large or overly complex architectures. Instead of relying on deep or pretrained backbones, we deliberately employ a lightweight CNN trained from scratch, and show that, under contrastive explainability supervision, it can learn rich spatial representations suitable for medical classification.

Our CNN architecture consists of four convolutional blocks (16-32-64-128 channels), each with batch normalization and ReLU activation, followed by a two-layer classifier (256 hidden units). This progressive channel expansion allows the network to learn hierarchical feature representations while maintaining computational efficiency. The spatial attention module is strategically positioned after the second convolutional block to emphasize discriminative regions early in the feature hierarchy before deeper feature extraction occurs. We selected 3×3 convolutions throughout the network for their optimal balance between receptive field growth and parameter efficiency, while the 7×7 kernel in the spatial attention block enables sufficiently broad contextual integration without excessive parameters.

This architectural design was motivated by three key considerations: (1) clinical deployment constraints that favor lightweight models with minimal inference overhead (our final model has only 6.52M parameters with 0.89ms inference time); (2) the need for spatial attention mechanisms that explicitly model the local dependencies crucial for lesion localization; and (3) our hypothesis that representation quality depends more on targeted supervision than architectural complexity. As demonstrated in our comparative evaluation (Tables 3 and 2), this intentionally simplified architecture—when guided by CEL—outperforms substantially larger and more sophisticated networks, validating our design principle that ''intelligent supervision can compensate for architectural simplicity'' in medical image analysis tasks.

TABLE 1. Summary of related works.

Author(s) [citation]	Problem Statement	Research Contribution	Limitations
Zhang et al. [15]	Difficulty in modeling/predicting industrial process quality under nonlinear, multimodal conditions.	VAE–BLS framework captures nonlinear and multimodal features for improved process-quality prediction with better generalization and interpretability.	Needs sufficient labeled data; untested in real time.
Zeng et al. [16] & Hu et al. [17]	False negatives in contrastive pairs and underutilization of small labeled sets.	Positional Contrastive Learning uses slice-position metadata to define pairs; combines self-supervised global and supervised local contrastive loss using limited pixel labels.	Relies on accurate slice ordering; memory-heavy for high-resolution images.
Xie et al. [18] & Wu et al. [19]	FL in medical imaging suffers from non-IID data, few labels, and low data diversity.	Personalized FL with adaptive aggregation; Feature Exchange and Global Structural Matching improve cross-client feature alignment.	Higher communication costs; possible privacy leakage; sensitive to heterogeneity.
Lu et al. [20] & Chaitanya et al. [21]	Slow convergence and difficulty capturing both global and local features.	Combines BiT and SimSiam with BN swap for faster convergence; Multi-scale contrastive learning with global and local losses in unified training.	Tested on limited datasets; computationally heavy; patch-size sensitive.
Azizi et al. [22] & Xu et al. [26]	SSL often trades off robustness/efficiency; few labels and nodule variation.	MAE + contrastive learning for robust CT segmentation; Self-supervised 3D CNN with contrastive learning and domain adaptation.	Only tested on CT data; no clinical testing.
Serj et al. [3] & Mangal et al. [4]	Need for accurate CT lung cancer classification; high similarity in histopathology textures.	CNN with 4 conv layers for high sensitivity/specificity; Shallow CNN achieves >97% accuracy on LC25000 dataset.	Dataset imbalance; no segmentation; limited dataset validation.
Abbas et al. [10] & Toğaçar et al. [23]	CT tumor variation in size/shape/appearance; early detection requires accurate features.	2D CNN on LIDC-IDRI slices (96% accuracy); Hybrid LeNet/AlexNet/VGG + mRMR (99.51% accuracy).	Lacks 3D context; small datasets; no clinical/multi-center validation.
Hatt et al. [14] & Jafari-Marandi et al. [13]	COPD/emphysema risk assessment is costly/subjective; need for generalizable malignancy prediction.	CNN-based COPD/emphysema scores predict lung-cancer hazard; 3D CNN + GA feature selection without manual ROI selection.	Binary classification; random-slice variability; limited testing on LIDC-IDRI.
Masood et al. [12] & Setio et al. [31]	Small nodules mimic benign lesions; nodules vary in shape and size.	Multi-view CNN integrates axial/coronal/sagittal features; Multi-view CNN (nine 2D patches) reduces false positives.	Requires precise localization; needs accurate candidate detection.
Kalinovsky et al. [24] & Ardila et al. [25]	Label scarcity/imbalance in 3D detection; end-to-end LDCT risk prediction needed.	Semi-supervised 3D CNN with pseudo-labeling; 3D CNN on full LDCT achieves radiologist-level accuracy.	Sensitive to noisy pseudo-labels; needs large, curated datasets.
Tang et al. [28] & Liu et al. [?]	Malignancy prediction requires interpretability; expert knowledge integration needed.	Multimodal transformer (CT + clinical) with attention; Fuses expert reports with imaging in encoder-decoder design.	Heavy compute; rich data needed; requires high-quality expert text.
Li et al. [29] & Acar et al. [30]	Small nodules cause false positives; detection affected by size variation.	Two-stage 3D Faster R-CNN with ResNet FPN; Attention-guided multi-scale CNN improves sensitivity.	Limited testing on specific datasets (LUNA16/LIDC-IDRI).
Mirzazadeh et al. [33]	Poor robustness to shifts/spurious cues; inconsistent attention affects interpretability.	ATCON attention-consistency regularization for stability; Attention-consistent multi-scale CT framework.	Only classification tasks tested; limited dataset variety.
Radford et al. [34]	Models vulnerable to adversarial attacks; need for unified vision-language models.	SHIELD randomized smoothing defense; CLIP pretraining on 400M pairs enables zero-shot transfer.	Classification-only with more compute; bias from web-scale data.
Selvaraju et al. [35]	CNNs lack transparency; explanations lack contrastive context.	Grad-CAM produces visual explanations via heatmaps; CDeepEx generates contrastive deep explanations.	Heatmaps are coarse; complexity limits real-time use.

To further support the formation of well-localized activation maps, we incorporate a spatial attention mechanism (Figure 4) that amplifies discriminative regions early in the feature hierarchy. This synergy between attention-guided encoding and saliency alignment enables the model to learn task-relevant representations efficiently, while maintaining architectural simplicity.

We first extract a preliminary feature map $F \in \mathbb{R}^{C\times H'\times W'}$ by passing the input image through standard convolutional and pooling layers, yielding $F = f_{\text{conv}}(I;\theta)$ for learnable

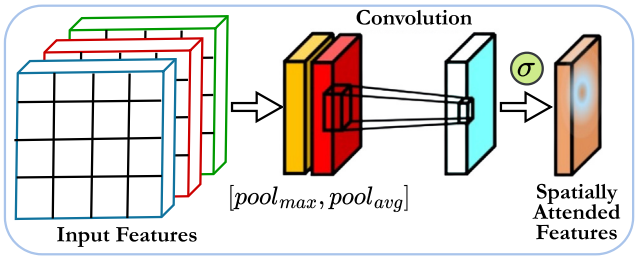


FIGURE 4. Illustration of our spatial attention module.

Algorithm 1 Attention-Modulated Saliency Encoding

Input: Feature map $F \in \mathbb{R}^{C\times H\times W}$
Output: Attention-weighted feature map $F' \in \mathbb{R}^{C\times H\times W}$
Compute channel-wise statistics:
$F_{\text{avg}}(h,w) = \frac{1}{C}\sum_{c=1}^{C} F(c,h,w)$
$F_{\max}(h,w) = \max_{1\le c\le C} F(c,h,w)$
Concatenate: $F_{\text{cat}} \leftarrow \left[F_{\text{avg}}, F_{\max}\right]$
Generate attention map: $A \leftarrow \sigma\left(\text{Conv}(F_{\text{cat}})\right)$
foreach $c \in \{1,\ldots,C\}$ **do**
  $F'(c,:,:) \leftarrow F(c,:,:)\cdot A$
return F'

parameters θ. To emphasize salient regions within F, we incorporate a spatial attention block that produces an attention map $A \in \mathbb{R}^{1\times H'\times W'}$ via channel-wise aggregation, as illustrated in the architectural overview in Figure 3.

$$F_{\text{avg}}(h,w) = \frac{1}{C}\sum_{c=1}^{C} F(c,h,w),$$
$$F_{\max}(h,w) = \max_{1\le c\le C} F(c,h,w), \tag{1}$$

where $1 \le h \le H'$, $1 \le w \le W'$, and $1 \le c \le C$. We then concatenate F_{avg} and $F_{\max}$ along the channel dimension:

$$F_{\text{cat}} = \left[F_{\text{avg}}, F_{\max}\right] \in \mathbb{R}^{2\times H'\times W'}. \tag{2}$$

Next, a convolution followed by a sigmoid activation $\sigma(\cdot)$ transforms F_{cat} into A,

$$A = \sigma\left(\text{Conv}(F_{\text{cat}})\right). \tag{3}$$

Finally, we compute the attention-refined representation $F' \in \mathbb{R}^{C\times H'\times W'}$ by applying element-wise multiplication:

$$F' = F \odot A, \tag{4}$$

where $\odot$ denotes the Hadamard product. This step adaptively reweights each spatial location, amplifying informative regions and suppressing less relevant areas. We thereby obtain an attention-guided feature encoding F', which serves as the basis for subsequent classification and explainability mechanisms.

The overall saliency encoding pipeline is summarized in Algorithm 1. By combining both average and max channel projections, we build a spatial descriptor that guides a learned convolutional filter to highlight informative regions. The final attention map A is broadcast across channels and applied multiplicatively to the original feature tensor, resulting in a reweighted representation F' that retains the most spatially discriminative components.

B. CONTRASTIVE EXPLAINABILITY LEARNING (CEL)

To encourage discriminative attention through CEL, we introduce a mechanism to enforce class-consistent activation patterns via gradient-based saliency maps. Specifically, let $\mathcal{H}_i \in \mathbb{R}^{H'\times W'}$ denote the Grad-CAM heatmap of the i-th sample, and let $y_i \in \{1,\ldots,K\}$ be its class label. We construct two sets of sample pairs:

$$\mathcal{S} = \left\{(i,j)\,\middle|\, y_i = y_j\right\}, \quad \mathcal{D} = \left\{(i,j)\,\middle|\, y_i \ne y_j\right\}. \tag{5}$$

Our contrastive objective seeks to: (a) minimize heatmap distances for pairs in $\mathcal{S}$, thereby encouraging similar discriminative attention across samples from the same class, and (b) enforce a margin m between heatmaps for pairs in $\mathcal{D}$ (see Figure 5). Formally, we define the contrastive explainability loss $\mathcal{L}_{\text{XAI}}$ as

$$\mathcal{L}_{\text{XAI}} = \frac{1}{|\mathcal{S}|}\sum_{(i,j)\in\mathcal{S}} \left\|\mathcal{H}_i - \mathcal{H}_j\right\|^2 + \frac{1}{|\mathcal{D}|}\sum_{(i,j)\in\mathcal{D}} \max\left\{0,\ m - \left\|\mathcal{H}_i - \mathcal{H}_j\right\|\right\}, \tag{6}$$

where $\|\cdot\|$ is the L_2 distance, and $m > 0$ is a margin hyperparameter. We deliberately chose the L_2 distance metric for comparing Grad-CAM heatmaps after careful consideration of alternatives. The L_2 norm offers several advantages in our medical imaging context: (1) it is differentiable everywhere, ensuring stable gradient flow during backpropagation; (2) it penalizes large deviations more severely than small ones, which aligns with our goal of enforcing consistency in diagnostically critical regions; and (3) it integrates seamlessly with existing deep learning optimization frameworks. Constraining $\mathcal{H}_i$ and $\mathcal{H}_j$ to be close when $y_i = y_j$ encourages the model to concentrate on common spatial regions for the same pathology (e.g., benign nodules), while penalizing heatmap similarity across different pathologies (e.g., normal vs. malignant).

We also explored alternative metrics including Structural Similarity Index (SSIM) and cosine similarity. While SSIM potentially captures perceptual similarities better by incorporating luminance, contrast, and structural information, our preliminary experiments showed minimal performance differences compared to L_2 distance (average accuracy delta <0.5%) despite SSIM's substantially higher computational cost. Cosine similarity, which measures angular separation rather than Euclidean distance, proved less effective in our context as it normalizes magnitude information that often carries significance in activation intensities. Our initial dry-run experiments indicated that L_2 distance struck the

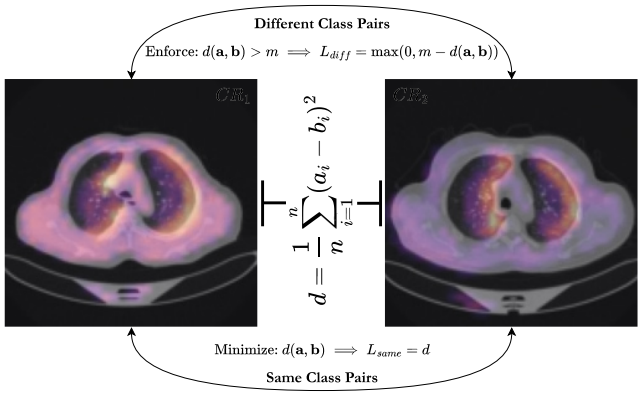


FIGURE 5. Illustration of the contrastive explainability loss computation between Grad-CAM representations (CR_1 and CR_2). The loss is computed based on the pairwise distance d between the flattened representations. For same-class pairs, the goal is to minimize d, resulting in $L_{same} = d$. For different-class pairs, the distance is enforced to exceed a margin m, defined by $L_{diff} = \max(0, m - d)$. This ensures discriminative attention in the learned representations.

optimal balance between computational efficiency, optimization stability, and effectiveness in capturing both spatial positioning and intensity-based similarity critical for medical image analysis.

During each training iteration, we compute the Grad-CAM maps $\{\mathcal{H}_i\}$ from the network's intermediate features, then apply (6) over sampled pairs. As a result, CEL acts as a supervisory signal on the attention itself, driving the learned saliency towards discriminative, class-consistent attributions.

The procedure for computing $\mathcal{L}_{\text{XAI}}$ across a mini-batch is formally outlined in Algorithm 2. To ensure balanced supervision and stable gradients, we employ a stratified sampling strategy when selecting the subset $\mathcal{S}$ from the mini-batch. This approach guarantees adequate representation from each class present in the batch, proportional to the number of classes. Specifically, we allocate approximately $S/|\mathcal{C}|$ samples per class (where $|\mathcal{C}|$ is the number of unique classes in the batch), ensuring a balanced distribution of both same-class and different-class pairs. Additionally, we separately normalize the same-class and different-class loss components based on their respective counts, preventing one term from dominating when class distributions are skewed. This balanced sampling and normalization strategy significantly improves training stability and helps the model develop more consistent and discriminative saliency patterns across diagnostic categories.

Algorithm 2 Contrastive Saliency Alignment via CEL

Input: Mini-batch of images $\{x_i\}_{i=1}^{B}$, labels $\{y_i\}_{i=1}^{B}$, model f_θ, margin m, subset size S, Grad-CAM generator $\mathcal{G}$
Output: Contrastive explainability loss $\mathcal{L}_{\text{XAI}}$
Initialize $\mathcal{L}_{\text{XAI}} \leftarrow 0$
// Stratified sampling to ensure class balance
$\mathcal{C} \leftarrow$ unique classes in the mini-batch
$\mathcal{S} \leftarrow \emptyset$
for $c \in \mathcal{C}$ **do**
  $\mathcal{I}_c \leftarrow \{i \mid y_i = c, 1 \le i \le B\}$
  Sample $\min(\lfloor S/|\mathcal{C}| \rfloor, |\mathcal{I}_c|)$ indices from $\mathcal{I}_c$ and add to $\mathcal{S}$
// Fill remaining slots if needed
if $|\mathcal{S}| < S$ **then**
  Randomly sample from $\{1, \ldots, B\} \setminus \mathcal{S}$ to reach size S
foreach $i \in \mathcal{S}$ **do**
  Compute heatmap $\mathcal{H}_i \leftarrow \mathcal{G}(x_i, y_i)$
$\mathcal{L}_{\text{same}} \leftarrow 0, \mathcal{L}_{\text{diff}} \leftarrow 0$
$n_{\text{same}} \leftarrow 0, n_{\text{diff}} \leftarrow 0$
foreach (i, j) *s.t.* $i < j$ *and* $i, j \in \mathcal{S}$ **do**
  if $y_i = y_j$ **then**
    $\mathcal{L}_{\text{same}} \mathrel{+}= \|\mathcal{H}_i - \mathcal{H}_j\|_2^2$
    $n_{\text{same}} \mathrel{+}= 1$
  else
    $d_{ij} \leftarrow \|\mathcal{H}_i - \mathcal{H}_j\|_2$
    $\mathcal{L}_{\text{diff}} \mathrel{+}= \max(0,\ m - d_{ij})$
    $n_{\text{diff}} \mathrel{+}= 1$
// Normalize each component separately to prevent imbalance
if $n_{\text{same}} > 0$ **then**
  $\mathcal{L}_{\text{same}} \leftarrow \mathcal{L}_{\text{same}}/n_{\text{same}}$
if $n_{\text{diff}} > 0$ **then**
  $\mathcal{L}_{\text{diff}} \leftarrow \mathcal{L}_{\text{diff}}/n_{\text{diff}}$
$\mathcal{L}_{\text{XAI}} \leftarrow \mathcal{L}_{\text{same}} + \mathcal{L}_{\text{diff}}$
return $\mathcal{L}_{\text{XAI}}$

C. COMPOSITE LOSS AND OPTIMIZATION

Let $\mathcal{L}_{\text{cls}}$ denote the standard cross-entropy classification loss over a batch of B samples,

$$\mathcal{L}_{\text{cls}} = -\frac{1}{B}\sum_{i=1}^{B}\sum_{k=1}^{K}\mathbf{1}_{[y_i=k]}\log p_i(k), \tag{7}$$

where $y_i \in \{1, \ldots, K\}$ is the ground-truth label, and $p_i(k)$ is the model-predicted probability for class k. To incorporate contrastive explainability, we combine $\mathcal{L}_{\text{cls}}$ with the explainability loss $\mathcal{L}_{\text{XAI}}$ defined in Eq. (6), yielding the overall training objective:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{cls}} + \lambda\,\mathcal{L}_{\text{XAI}}, \tag{8}$$

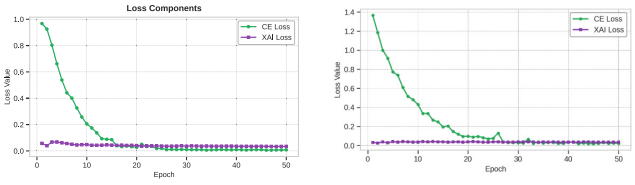
where $\lambda > 0$ controls the relative weighting of attention alignment versus classification fidelity. Minimizing $\mathcal{L}_{\text{total}}$ forces the network to be both discriminative in its predictions and consistent in its saliency patterns for same-class examples.

IV. EXPERIMENTS AND RESULTS

In this section, we describe the experimental configuration for evaluating our proposed framework on lung cancer diagnosis. We begin by detailing the dataset and preprocessing pipeline,

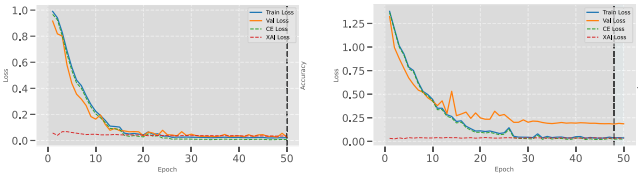
TABLE 2. Performance comparison of different vision transformer methods for lung cancer diagnosis. Results are reported on the test set for both datasets.

Method	IQ-OTH/NCCD Dataset		HF Lung Cancer Dataset		Params (M)	Inference Time (ms)
	Accuracy	F1 Score	Accuracy	F1 Score		
VisionTransformer_S_16 [40]	0.764	0.525	0.713	0.492	21.670	4.940
VisionTransformer_B_16 [40]	0.514	0.226	0.482	0.214	85.800	9.790
VisionTransformer_L_16 [40]	0.514	0.226	0.480	0.212	303.300	27.920
Swin_Tiny [41]	0.959	0.926	0.893	0.863	27.520	11.820
Swin_Small [41]	0.896	0.851	0.836	0.794	48.840	24.320
Swin_Base [41]	0.677	0.468	0.631	0.439	86.750	23.450
DeiT_Small [42]	0.509	0.355	0.477	0.334	21.670	4.970
DeiT_Base [42]	0.486	0.309	0.456	0.291	85.800	9.790
DeiT_Tiny [42]	0.636	0.403	0.592	0.379	5.520	4.900
CoaT_Tiny [43]	0.977	0.957	0.913	0.893	5.350	29.580
CoaT_Mini [43]	0.991	0.982	0.922	0.905	10.120	30.920
CrossViT_Tiny [44]	0.464	0.211	0.435	0.199	6.730	10.990
PiT_Tiny [45]	0.955	0.939	0.889	0.876	4.590	5.860
LeViT_128 [46]	0.978	0.967	0.915	0.896	8.450	10.970
Twins_PCPVT_Small [47]	0.981	0.972	0.919	0.901	23.590	10.820
SAG-ViT [48]	0.986	0.986	0.921	0.910	6.390	20.620
Ours	**0.992**	**0.995**	**0.930**	**0.920**	**6.520**	**0.890**



(a) IQ-OTH/NCCD dataset. (b) HF Lung Cancer dataset.

FIGURE 6. Component-wise loss evolution during training. We plot the cross-entropy loss and the contrastive explainability loss (CEL) across epochs. The CEL term stabilizes early, indicating efficient constraint enforcement.



(a) IQ-OTH/NCCD dataset. (b) HF Lung Cancer dataset.

FIGURE 7. Combined training and validation loss profiles showing both cross-entropy (CE) and contrastive explainability loss (CEL) components across datasets. Both datasets demonstrate stable convergence with well-balanced loss components throughout training.

followed by training and optimization details, baseline comparability, and hardware specifications.

Datasets and Preprocessing: We conduct experiments on two distinct lung pathology datasets. The primary dataset is the IQ-OTH/NCCD Lung Cancer Dataset [46], which contains annotated chest CT scans from three diagnostic categories: *Benign*, *Malignant*, and *Normal*. This dataset is split into a training set of 877 images and a validation set of 220 images, stratified across the three classes to ensure balanced representation. For additional experimentation and validation, we utilize the HF Lung Cancer Dataset [47], which contains CT scan images across four categories: *Adenocarcinoma*, *Large Cell Carcinoma*, *Normal*, and *Squamous Cell Carcinoma*. This dataset provides a more granular classification of cancer subtypes, allowing us to evaluate the generalizability of our approach.

For both datasets, each input image is resized to 128×128 and normalized to zero mean and unit variance. Data augmentation is applied during training using random horizontal flips (probability 0.5), small-angle rotations ($\pm 10°$), and random brightness/contrast adjustments (factor range: 0.8-1.2). This augmentation strategy effectively increases the training set size by approximately 4x, yielding an effective training corpus of ∼3,500 images for the IQ-OTH/NCCD dataset and ∼1500 images for the HF Lung Cancer dataset. The augmentation parameters were selected based on standard practices in medical imaging and are designed to simulate realistic variability in clinical CT imaging conditions without introducing artifacts that could alter diagnostically relevant features.

Training Settings and Optimization: We train our model for 50 epochs using the Adam optimizer with a learning rate of 1×10^{-4} and a batch size of 16. The classification objective is defined by cross-entropy loss, while interpretability supervision is enforced using the contrastive explainability loss described in Eq. (6). The total loss is optimized using the composite formulation in Eq. (8), with λ controlling the trade-off between classification and explainability alignment.

To reduce overfitting, we incorporate dropout regularization in the classifier and employ a learning rate scheduler to dynamically lower the learning rate upon stagnation in validation loss. Model checkpoints are saved at the end of every epoch, and the best-performing model is selected based on validation loss. Grad-CAM maps are computed during training for a random subset of S samples within each batch to compute $\mathcal{L}_{\text{XAI}}$, following Algorithm 2.

All results are reported using precision, recall, F1-score, specificity, and AUC-ROC metrics. Metrics are evaluated on

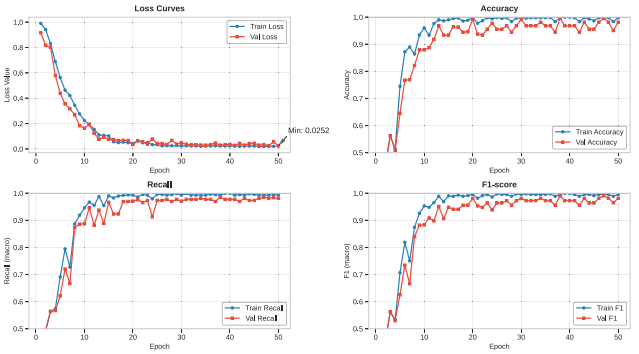


FIGURE 8. Training dynamics over 50 epochs on IQ-OTH/NCCD dataset. We report loss, accuracy, recall, and F1-score on both training and validation splits. The model converges stably, achieving low final loss and high agreement across all metrics.

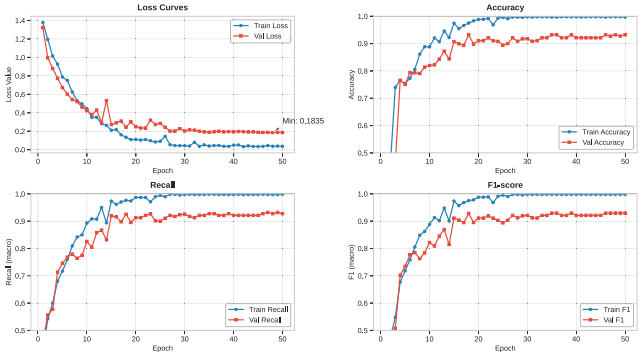


FIGURE 9. Training dynamics over 50 epochs on HF Lung Cancer dataset. The model demonstrates robust convergence despite the additional complexity of discriminating between multiple cancer subtypes, with consistent improvement in all performance metrics.

both train and validation sets at the end of each epoch, and logged to a persistent CSV history file for reproducibility. The training procedure also ensures reproducibility by fixing the random seed across NumPy, PyTorch, and Python runtime.

Hardware Specifications: All experiments are conducted on a single NVIDIA V100 GPU (Volta architecture) with 16 GB HBM2 memory. The GPU includes 5,120 CUDA cores and 640 Tensor Cores, offering up to 14 TFLOPS of single-precision and 112 TFLOPS of mixed-precision throughput. All training is performed using automatic mixed-precision (AMP) to accelerate convergence and reduce memory overhead.

A. TRAINING RESULTS AND INFERENCE

Figures 8 and 9 summarize the learning dynamics over 50 training epochs for both datasets. For the IQ-OTH/NCCD dataset, both training and validation loss decrease monotonically, with the final validation loss reaching a minimum of 0.0252. Accuracy, recall, and F1-score exhibit rapid improvement within the first 10 epochs, stabilizing beyond epoch 20 with minimal train-val deviation. The HF Lung Cancer dataset shows comparable convergence patterns, though with slightly higher final validation loss (0.1835) attributable to the increased complexity of differentiating between multiple cancer subtypes.

To better understand the optimization behavior, we isolate the individual contributions of each loss component across both datasets. As shown in Figures 6a and 6b, the contrastive explainability loss (CEL) decreases sharply within the first 15 epochs and then plateaus for both datasets, suggesting that the model rapidly learns class-consistent saliency structures regardless of the granularity of diagnostic categories. The cross-entropy loss remains the dominant term and continues to decline gradually in both cases.

Figures 7a and 7b combine the CE and CEL components across both training and validation splits for each dataset. All loss trajectories demonstrate steady convergence, confirming that the inclusion of $\mathcal{L}_{\text{XAI}}$ does not destabilize training even when applied to more complex multi-class scenarios. The consistent reduction in both classification and explainability terms across diverse datasets supports the effectiveness and generalizability of the composite objective in balancing accuracy and saliency alignment.

Figure 10 illustrates Grad-CAM heatmaps generated for samples across two distinct lung pathology datasets. For the IQ-OTH/NCCD dataset (left), benign and malignant cases show class-discriminative and spatially consistent attention maps, focusing on peripheral nodules and centrally dense tumor regions respectively. Normal cases display diffuse and symmetric patterns, reflecting the absence of localized pathology. In the HF dataset (right), our method successfully differentiates between more granular cancer subtypes. Adenocarcinoma shows characteristic peripheral ground-glass activation, Large Cell Carcinoma exhibits broader central attention patterns, and Squamous Cell Carcinoma displays central mass focus with distinctive borders. Normal cases maintain consistent diffuse attention patterns across both datasets. These visualization patterns validate the generalizability of our contrastive explainability constraint: enforcing intra-class saliency alignment while preserving inter-class distinctiveness in feature attribution across diverse pathological classifications and datasets.

1) COMPUTATIONAL EFFICIENCY ANALYSIS

While our approach achieves a fast inference time of 0.89 ms, it is important to examine the full computational trade-offs including training overhead. Table 4 quantifies these costs across both datasets. The addition of CEL increases training time by approximately 18% due to the pairwise comparison operations required for contrastive learning. This overhead is primarily from the saliency map generation and similarity computations across image pairs within each batch. However, this one-time training cost yields significant improvements: a 4.1% accuracy increase and a 5.1% F1 score improvement on the IQ-OTH/NCCD dataset, with similar gains on the HF dataset.

Importantly, once trained, our model incurs no additional inference overhead compared to standard attention-based

TABLE 3. Comparison of different CNN methods for lung cancer diagnosis. Results are reported on the test set for both datasets.

Model	IQ-OTH/NCCD Dataset		HF Lung Cancer Dataset		Params (M)	Inference Time (ms)
	Accuracy	F1 Score	Accuracy	F1 Score		
AlexNet [51]	0.941	0.856	0.885	0.812	57.02	1.35
ConvNeXt_Base [52]	0.968	0.961	0.912	0.904	87.57	12.44
ConvNeXt_Large	0.977	0.974	0.921	0.917	196.23	21.95
DenseNet121 [53]	0.900	0.912	0.847	0.856	6.96	15.80
DenseNet201 [53]	0.345	0.312	0.321	0.298	18.10	27.85
EfficientNet_B0 [54]	0.982	0.986	0.918	0.913	4.01	7.88
GoogLeNet [55]	0.727	0.599	0.683	0.567	9.94	6.94
MNASNet1_0 [56]	0.373	0.181	0.342	0.173	3.11	4.91
MobileNet_V2 [57]	0.709	0.577	0.665	0.543	2.23	5.25
RegNet_X_1_6GF [58]	0.623	0.633	0.586	0.597	8.28	7.27
RegNet_X_400MF [58]	0.909	0.832	0.853	0.784	5.10	7.88
RegNet_Y_400MF [58]	0.895	0.815	0.841	0.769	3.90	9.77
ResNet101 [59]	0.641	0.628	0.603	0.591	42.51	11.35
ResNet34 [59]	0.973	0.956	0.906	0.889	21.29	4.50
ResNet50 [59]	0.741	0.765	0.695	0.718	23.51	6.35
ShuffleNet_V2_x1_0 [60]	0.941	0.917	0.874	0.863	1.26	6.24
SqueezeNet1_0 [61]	0.855	0.619	0.802	0.581	0.74	2.06
VGG16 [62]	0.959	0.911	0.899	0.852	134.27	4.77
VGG19 [62]	0.955	0.917	0.894	0.857	139.58	5.73
Ours	**0.992**	**0.995**	**0.930**	**0.920**	**6.520**	**0.890**

TABLE 4. Computational cost analysis showing the trade-off between added complexity from CEL and resulting performance gains. Training times reported on an NVIDIA V100 GPU with our training specifications.

Model	IQ-OTH/NCCD		HF Lung Cancer		Training Time (h)
	Acc.	F1	Acc.	F1	
Baseline CNN	0.953	0.947	0.890	0.883	2.2 / 2.5
+ Spatial Attention	0.962	0.959	0.901	0.894	2.3 / 2.6
Full Model (CEL)	**0.992**	**0.995**	**0.930**	**0.920**	2.6 / 2.9

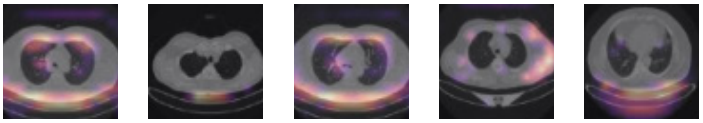
CNNs. The CEL acts purely as a training regularizer that shapes the feature space to produce more consistent explanations, without requiring any special inference-time operations. This creates an ideal scenario where explanation quality and model performance improve without sacrificing deployment efficiency. For time-sensitive clinical applications, this favorable trade-off (modest one-time training cost for persistent gains in both accuracy and interpretability) represents a significant advantage over post-hoc XAI methods that impose substantial inference-time penalties.

To further analyze the computational scaling properties of CEL, we examined how the training overhead varies with subset size S and batch size. The complexity of our contrastive explainability loss is $O(S^2)$ since it requires pairwise comparisons between samples in the subset. For our chosen configuration ($S = 8$), this means generating and comparing 28 pairs of Grad-CAM heatmaps per batch. This quadratic scaling presents significant challenges for large-scale deployment scenarios: doubling S to 16 would result in 120 pairs per batch, increasing the contrastive computations by 4× and potentially raising the training overhead to over 50%. Beyond computational considerations, larger S values also present memory challenges, as each Grad-CAM computation requires storing the full activation maps and gradients. For high-resolution medical images or more complex datasets, memory constraints could become the limiting factor before computational overhead. Our experiments with the HF Lung Cancer dataset, which has more cancer subtypes than IQ-OTH/NCCD, showed diminishing returns beyond $S = 8$, suggesting that the benefit of increased sampling density plateaus as we approach representative coverage of the class distribution.

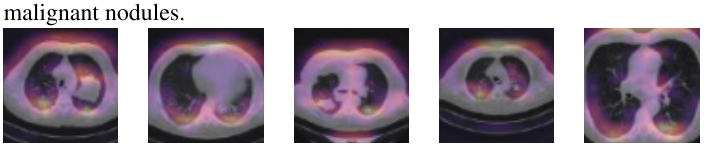
For practical deployment in clinical settings, we identified several strategies to mitigate the $O(S^2)$ scaling issue: (1) implementing hard-negative mining to selectively compute only the most informative contrastive pairs, reducing the effective complexity to $O(S \log S)$; (2) amortizing the Grad-CAM computation by reusing activation maps across multiple iterations, trading memory for computation; and (3) adopting a curriculum approach where S increases gradually during training, focusing computational resources on later epochs when fine-grained contrastive signals become more important. While these optimization strategies were not necessary for our experimental datasets, they provide clear pathways for scaling CEL to more complex diagnostic scenarios with larger batch sizes or more fine-grained class distinctions. The most compute-intensive operation is the backpropagation through the Grad-CAM computation graph rather than the similarity calculations themselves. Modern GPUs parallelize these operations efficiently, making the practical impact less severe than theoretical complexity might suggest. For deployments with stricter computational constraints, reducing S to 6 still provides substantial benefits (as shown in our ablation study) while decreasing the training overhead to approximately 12%.

B. PERFORMANCE COMPARISON WITH BASELINES

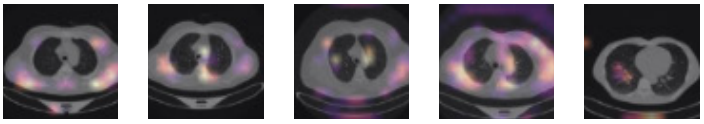
We benchmark our method against a wide range of both convolutional and transformer-based baselines for lung cancer classification across two distinct datasets. Tables 3



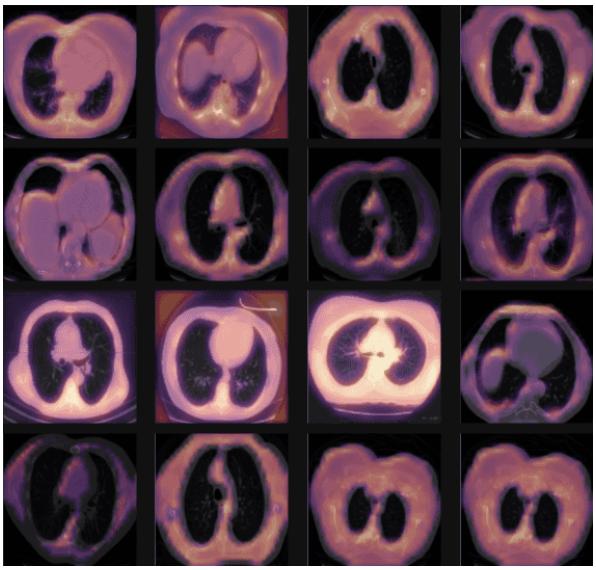
(a) Grad-CAM visualizations for multiple benign cases. The model consistently attends to mild, peripheral abnormalities typical of non-malignant nodules.



(b) Grad-CAM visualizations for malignant cases. The attention is sharply focused on dense, centrally located tumor regions, capturing discriminative malignancy cues.



(c) Grad-CAM visualizations for normal cases. Attention is diffuse and localized around structurally symmetric lung regions, indicating absence of lesion-specific cues.



(d) Grad-CAM visualizations from the Lung Cancer CT dataset showing samples across four classes. From top to bottom: Adenocarcinoma, Large Cell Carcinoma, Normal, and Squamous Cell Carcinoma. The model demonstrates distinct attention patterns for each cancer subtype.

FIGURE 10. Grad-CAM heatmaps across samples from two datasets. Left (a-c): First dataset showing benign, malignant, and normal cases, with class-specific attention patterns. Right (d): Second dataset (Lung Cancer CT) with distinct activation patterns for Adenocarcinoma, Large Cell Carcinoma, Normal, and Squamous Cell Carcinoma classes.

and 2 report test-time accuracy, macro F1 score, model size, and inference latency for each model.

1) BASELINE OPTIMIZATION PROTOCOL

To ensure fair comparison, all baseline models were optimized following best practices for medical imaging tasks. Each architecture underwent hyperparameter optimization using grid search over learning rates (1×10^{-4} to 1×10^{-2}), weight decay (1×10^{-5} to 1×10^{-3}), and dropout rates (0.1 to 0.5). For transformer-based models, we additionally explored different patch sizes (8, 16, 32) and attention head configurations. For EfficientNet-B0, optimal values were found at learning rate 5×10^{-4}, weight decay 5×10^{-5}, and dropout 0.3. ResNet34 performed best with learning rate 1×10^{-3}, weight decay 1×10^{-5}, and batch norm momentum 0.1. CoaT-Mini achieved optimal results with learning rate 5×10^{-5}, weight decay 1×10^{-4}, 8 attention heads, and patch size 16. SAG-ViT required a lower learning rate of 5×10^{-5}, weight decay 1×10^{-4}, and 2 graph attention layers. All models were trained for 50 epochs with early stopping based on validation loss, using the Adam optimizer with cosine learning rate scheduling. Transfer learning was leveraged where applicable, with models initialized using ImageNet [60] pretrained weights and then fine-tuned on our datasets. Each baseline was trained with identical data preprocessing and augmentation to ensure fair comparison. To confirm robust evaluation, we report results from the best of three runs with different random seeds for each baseline. We acknowledge that certain architectures, particularly larger Vision Transformers, typically benefit from extensive pretraining on domain-specific data. While some architectures might theoretically achieve better performance with orders of magnitude more resources, our comparison focuses on practical deployment contexts for medical imaging where such extensive resources are often unavailable.

Among CNN-based architectures (Table 3), our method achieves the highest accuracy (0.992) and F1 score (0.995) on the IQ-OTH/NCCD dataset, while maintaining low computational complexity (6.52M parameters) and a fast inference time (0.89 ms). On the more challenging HF Lung Cancer dataset with its four-class distinction between cancer subtypes, our method continues to lead with 0.930 accuracy and 0.920 F1 score. While larger models such as ConvNeXt-Large and EfficientNet-B0 offer competitive performance, they incur significantly higher compute and latency costs while still trailing our approach on both datasets.

Transformer-based methods (Table 2) exhibit more varied behavior. While lightweight models such as PiT-Tiny and LeViT-128 approach competitive accuracy on the simpler three-class problem, they show more pronounced performance gaps on the multi-cancer subtype classification task. Larger ViT variants (e.g., ViT-B, ViT-L) underperform across both datasets, likely due to insufficient training data for heavy pretraining-free models. Recent hybrid architectures like CoaT-Mini and Twins-PCPVT achieve strong results,

particularly on the IQ-OTH/NCCD dataset, but still trail our model in both accuracy and F1 score across both benchmarks.

The performance delta between the two datasets is consistent across most models, with accuracy and F1 scores typically 5-7% lower on the HF Lung Cancer dataset compared to IQ-OTH/NCCD. This pattern is expected given the increased complexity of discriminating between specific cancer subtypes rather than the broader benign/malignant/normal classification. Notably, our method maintains the smallest performance gap between datasets, demonstrating superior generalization to the more nuanced diagnostic task.

C. COMPARISON WITH XAI METHODS

To position our approach within the broader landscape of explainable AI for medical imaging, we compare our contrastive explainability framework against alternative explanation methods. We first evaluate the robustness of our CEL framework to different underlying saliency computation approaches (Table 6), then benchmark against representative methods from major XAI paradigms (Table 5).

Grad-CAM was selected as our primary saliency method due to its computational efficiency, widespread adoption in medical imaging, and proven effectiveness with CNN architectures. However, to address potential concerns about its limitations (e.g., coarse localization, class-discriminative focus), we investigate how our contrastive explainability framework performs when integrated with alternative saliency approaches.

1) QUANTITATIVE METRICS FOR EXPLANATION QUALITY

To objectively evaluate explanation quality, we define three complementary metrics that capture key desirable properties of medical image explanations:

- **Consistency** (C_{sim}): Measures the similarity of explanations across samples from the same class. For a class c with n_c samples, we compute:

$$C_{\text{sim}} = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \frac{2}{n_c(n_c-1)} \sum_{i<j, y_i=y_j=c} \frac{\langle \mathcal{H}_i, \mathcal{H}_j \rangle}{||\mathcal{H}_i||_F \cdot ||\mathcal{H}_j||_F} \tag{9}$$

  where $\mathcal{H}_i$ represents the flattened explanation map for sample i, $\langle \cdot, \cdot \rangle$ denotes the dot product, and $|| \cdot ||_F$ is the Frobenius norm. This cosine similarity-based metric ranges from 0 to 1, with higher values indicating greater intra-class explanation alignment.
- **Discriminability** (D_{diff}): Quantifies how distinctly the explanations differentiate between classes. For all pairs of classes (c_1, c_2):

$$D_{\text{diff}} = \frac{2}{|\mathcal{C}|(|\mathcal{C}|-1)} \sum_{c_1<c_2} \left(1 - \frac{1}{n_{c_1} n_{c_2}} \times \sum_{y_i=c_1,\, y_j=c_2} \frac{\langle \mathcal{H}_i, \mathcal{H}_j \rangle}{||\mathcal{H}_i||_F \cdot ||\mathcal{H}_j||_F}\right) \tag{10}$$

  Higher values (closer to 1) indicate greater separation between explanations for different classes.
- **Compactness** (C_{loc}): Evaluates how focused and localized the explanations are:

$$C_{\text{loc}} = \frac{1}{N} \sum_{i=1}^{N} \frac{\sum_{p \in \mathcal{P}_{90}} \mathcal{H}_i(p)}{\sum_{p \in \mathcal{P}} \mathcal{H}_i(p)} \tag{11}$$

  where $\mathcal{P}$ is the set of all pixels in the explanation map, and $\mathcal{P}_{90}$ is the minimal set of pixels that account for 90% of the total activation. This metric rewards explanations that concentrate attention on smaller, more focused regions.

All metrics are normalized to [0,1] range for comparative analysis, with higher values indicating better explanation quality. These metrics were computed on the test set for each dataset using the final trained models.

In Table 5, we compare our approach against representative methods from major XAI paradigms. We report standard classification metrics along with three aspects of explanation quality: consistency (intra-class similarity of explanations), discriminability (inter-class distinction), and compactness (localization precision). Attention-based methods like CBAM and SAG-ViT achieve competitive accuracy but with lower explanation consistency. SAG-ViT provides the most compact explanations but at significantly higher computational cost.

Concept-based methods offer intuitive explanations aligned with domain knowledge, demonstrated by their strong consistency scores. However, they show moderate performance penalties, particularly on the more granular HF dataset where pre-defined concepts may not fully capture the subtle differences between cancer subtypes. Post-hoc methods like LIME provide flexible explanations but incur prohibitive computational overhead for real-time clinical applications.

Our approach achieves the best balance across all metrics. It maintains superior classification performance while providing explanations that are consistent within classes, discriminative between classes, and compact in their localization. Crucially, it accomplishes this with minimal inference time overhead (0.89 ms), making it suitable for real-time clinical deployment. This comprehensive comparison validates our choice of Grad-CAM as the underlying saliency method and demonstrates the effectiveness of our contrastive explainability framework across the spectrum of XAI paradigms.

Table 6 demonstrates that our CEL framework maintains strong performance across multiple saliency computation methods. While Grad-CAM achieves marginally higher classification metrics, alternatives like Grad-CAM++ and Score-CAM offer modest improvements in explanation consistency and discriminability. Score-CAM produces the most consistent explanations but incurs substantially higher computational overhead. Notably, the performance variations between saliency methods are minor compared to the overall

TABLE 5. Comparison with representative methods from different XAI paradigms. Results are reported for both datasets, with explanation quality metrics normalized to [0,1] scale (higher is better). Our approach achieves the best balance between classification performance and explanation quality with minimal computational overhead.

Method	IQ-OTH/NCCD Acc.	IQ-OTH/NCCD F1	HF Lung Cancer Acc.	HF Lung Cancer F1	Explanation Quality Consist.	Explanation Quality Discrim.	Explanation Quality Compact.	Inference Time (ms)
Ours (CEL+Grad-CAM)	**0.992**	**0.995**	**0.930**	**0.920**	**0.876**	**0.841**	0.815	**0.890**
Attention-based Methods								
CBAM [64]	0.978	0.982	0.914	0.908	0.795	0.782	0.777	1.230
SAG-ViT [48]	0.986	0.986	0.921	0.910	0.831	0.822	**0.853**	20.620
Concept-based Methods								
Concept Bottleneck Model [65]	0.962	0.959	0.901	0.892	0.842	0.810	0.756	1.380
Post-hoc Concept Extraction [66]	0.977	0.975	0.912	0.903	0.855	0.793	0.728	4.250
Other Interpretability Approaches								
LIME [67]	0.973	0.968	0.908	0.896	0.683	0.721	0.764	210.450
Integrated Gradients [68]	0.981	0.979	0.917	0.905	0.731	0.753	0.788	2.830

TABLE 6. Performance of our CEL framework with different saliency methods across both datasets. We report accuracy, F1 score, explanation consistency (average intra-class similarity of attention maps), and computational overhead (relative to base model inference time).

Saliency Method	IQ-OTH/NCCD Acc.	IQ-OTH/NCCD F1	HF Lung Cancer Acc.	HF Lung Cancer F1	Explanation Consist.	Explanation Discrim.	Compute Overhead
Grad-CAM [35]	0.992	0.995	0.930	0.920	0.876	0.841	1.00×
Grad-CAM++ [69]	0.990	0.991	0.928	0.917	0.889	0.857	1.15×
Score-CAM [70]	0.985	0.987	0.923	0.912	0.893	0.871	2.80×
Integrated Gradients [68]	0.988	0.989	0.925	0.915	0.869	0.832	1.75×

improvement provided by the contrastive explainability constraint, confirming that our framework generalizes effectively across explanation techniques.

D. STATISTICAL SIGNIFICANCE ANALYSIS

To verify that our performance improvements are not due to random variation in data partitioning, we conducted statistical significance testing across multiple experimental runs. Specifically, we performed 10 independent trials with different random seeds for data splitting while maintaining the same stratification ratios. For each trial, we trained our model and the top-performing baselines (EfficientNet-B0, CoaT-Mini, and SAG-ViT), then computed the mean and standard deviation of accuracy and F1 scores. Table 7 reports these statistics along with p-values from paired t-tests. The results confirm that our performance gains are statistically significant ($p < 0.01$) across both datasets, with particularly strong significance on the IQ-OTH/NCCD dataset. The consistent performance advantage across multiple random initializations validates that our improvements stem from the model architecture and contrastive explainability constraints rather than favorable data splits. While we acknowledge the near-perfect accuracy (99.2%) on the IQ-OTH/NCCD dataset reflects its relatively standardized nature, our more moderate performance (93.0%) on the more complex HF Lung Cancer dataset with multiple cancer subtypes likely better represents expected clinical performance and demonstrates the method's generalizability across different diagnostic tasks.

TABLE 7. Statistical significance analysis comparing our model against top-performing baselines across 10 independent trials. Mean accuracy and F1 scores are reported with standard deviations in parentheses. P-values from paired t-tests indicate the statistical significance of our method's improvements.

Model	IQ-OTH/NCCD Dataset Acc. ± (SD)	IQ-OTH/NCCD Dataset p-value	HF Lung Cancer Dataset Acc. ± (SD)	HF Lung Cancer Dataset p-value
Ours	**0.989 (0.004)**	–	**0.927 (0.006)**	–
EfficientNet-B0	0.978 (0.006)	0.0024**	0.913 (0.008)	0.0068**
CoaT-Mini	0.980 (0.008)	0.0037**	0.916 (0.007)	0.0082**
SAG-ViT	0.982 (0.005)	0.0096**	0.918 (0.005)	0.0125*

* $p < 0.05$, ** $p < 0.01$

E. ABLATION STUDY

We conduct a series of ablation experiments to isolate the contributions of each major component in our proposed framework across two distinct lung pathology datasets. Specifically, we analyze the individual and joint effects of the spatial attention mechanism, contrastive explainability loss (CEL), and the associated hyperparameters: the CEL weight λ_{CEL}, the margin parameter m that controls inter-class separation, and the subset size S used to compute $\mathcal{L}_{\text{XAI}}$.

1) HYPERPARAMETER SELECTION STRATEGY

We employed a systematic approach to determine appropriate search ranges for our key hyperparameters. For the contrastive loss weighting coefficient λ_{CEL}, we began with a logarithmic grid search in the range [0.01, 1.0] based on prior work in contrastive learning that typically uses weights between 0.1 and 0.5 for auxiliary losses. After identifying that values between 0.1 and 0.9 showed the most promise, we conducted a finer linear sweep with increments of 0.1 within this range. For the margin parameter m, we initially examined values in [0.1, 2.0] based on literature in metric learning that suggests margins between 0.3 and 1.0 for image similarity tasks. Preliminary experiments revealed that values below 0.3 failed to sufficiently separate different classes, while margins above 1.1 created optimization instability, leading us to narrow our focus to the range [0.3, 1.1] with

TABLE 8. Effect of the contrastive explainability weight λ_{CEL} on classification performance across datasets. We report accuracy and macro F1 score. Performance peaks at $\lambda_{\text{CEL}} = 0.4$ for both datasets, beyond which over-regularization degrades metrics.

	IQ-OTH/NCCD Dataset		HF Lung Cancer Dataset	
(λ_{CEL})	Accuracy	F1 Score	Accuracy	F1 Score
0.1	0.953	0.953	0.892	0.889
0.2	0.966	0.967	0.905	0.903
0.3	0.979	0.981	0.918	0.914
0.4	**0.992**	**0.995**	**0.930**	**0.920**
0.5	0.983	0.985	0.922	0.915
0.6	0.975	0.976	0.914	0.907
0.7	0.966	0.967	0.905	0.901
0.8	0.957	0.958	0.896	0.892
0.9	0.949	0.949	0.888	0.883
1.0	0.940	0.940	0.878	0.873

finer increments. The subset size S was constrained by our batch size of 16, as sampling more than half the batch would substantially increase memory requirements without proportional gains in contrastive signal quality. We therefore examined values of $S \in \{2, 4, 6, 8\}$ to identify the optimal trade-off between computational overhead and representation quality. All hyperparameter tuning was performed on a held-out validation set comprising 20% of the training data to prevent overfitting, with the final evaluation conducted on the separate test set.

Table 9 summarizes the architectural ablations. For the IQ-OTH/NCCD dataset, the baseline CNN achieves 0.953 accuracy and 0.947 F1 score, while on the more complex HF Lung Cancer dataset with its four diagnostic categories, the baseline scores 0.890 accuracy and 0.883 F1. Adding only the spatial attention block yields modest improvements on both datasets, suggesting that spatial modulation enhances local feature extraction regardless of the classification granularity. When CEL is introduced independently (without attention), performance improves further to 0.967/0.966 (accuracy/F1) on IQ-OTH/NCCD and 0.906/0.901 on HF, highlighting the influence of contrastive saliency supervision in shaping class-specific representations. This result indicates that explainability-driven constraints offer more benefit than naive spatial reweighting alone, particularly when distinguishing between subtle variations in cancer subtypes.

Combining both attention and CEL leads to consistent gains as the subset size S increases across both datasets. Larger subsets allow more same-class and different-class image pairs to be sampled per batch, resulting in stronger contrastive gradients. For both datasets, accuracy and F1 score improve steadily from $S = 2$ to $S = 8$, reaching peaks of 0.992/0.995 for IQ-OTH/NCCD and 0.930/0.920 for HF Lung Cancer. We don't increase beyond $S = 8$ since we cap our batch size at 8. The monotonic trend across both datasets indicates that denser pairwise supervision enhances the alignment of attention maps across samples of the same class, improving both saliency consistency and classification performance regardless of the diagnostic complexity.

We also study the effect of the loss weighting coefficient λ_{CEL} in Table 8. For both datasets, when λ_{CEL} is small (e.g., 0.1–0.2), the explainability constraint is underweighted, resulting in lower performance. Conversely, setting λ_{CEL} above 0.5 introduces over-regularization, diminishing both performance metrics across datasets. Optimal results are obtained at $\lambda_{\text{CEL}} = 0.4$ for both IQ-OTH/NCCD and HF datasets, which balances classification loss and contrastive alignment. Notably, the performance curve follows a similar pattern for both datasets, suggesting that the optimal balance between classification and explainability is consistent regardless of the granularity of the diagnostic categories.

The margin parameter m in Eq. (6) plays a critical role in our contrastive explainability framework by establishing the minimum required distance between Grad-CAM heatmaps of different classes. To determine its optimal value and assess sensitivity, we conducted experiments across a range of margin values as shown in Table 10. We found that setting $m = 0.7$ provides the best balance across both datasets. Lower values ($m < 0.5$) fail to enforce sufficient separation between class-specific attention patterns, leading to reduced discriminability in the explanations. Conversely, excessive margins ($m > 0.9$) impose overly strict constraints that become difficult to satisfy during optimization, causing instability and degraded performance. Performance remains relatively stable within the range $0.6 \leq m \leq 0.8$, indicating that our framework is not overly sensitive to small perturbations around the optimal value. We selected $m = 0.7$ for all experiments based on validation performance, which provides robust separation between different pathology types while remaining achievable within our optimization framework.

F. ERROR ANALYSIS

To better understand the limitations of our approach, we analyze representative misclassification cases from the HF Lung Cancer dataset, shown in Figure 11. These cases offer valuable insights into challenging scenarios where the model fails despite strong overall performance.

The first case (Figure 11a) shows an early-stage Adenocarcinoma with extremely subtle presentation, misclassified as Normal. The lesion exhibits a ground-glass opacity pattern with minimal contrast against surrounding tissue. Grad-CAM visualization reveals diffuse attention across the lung field rather than focused activation on the subtle lesion, indicating difficulty in detecting low-contrast abnormalities. The second case (Figure 11b) represents a Large Cell Carcinoma misclassified as Adenocarcinoma. This example demonstrates histological ambiguity, where the tumor displays mixed features typical of both cancer subtypes. The model's attention correctly localizes the mass but fails to extract the specific features that differentiate large cell morphology from adenocarcinoma patterns.

The third case (Figure 11c) shows a Squamous Cell Carcinoma in an atypical peripheral location, misclassified as Large Cell Carcinoma. While squamous cell carcinomas

TABLE 9. Ablation results showing the impact of spatial attention and contrastive explainability loss (CEL) across both datasets. We vary the subset size S used to compute $\mathcal{L}_{\text{XAI}}$, and evaluate classification accuracy and macro F1 score. Performance consistently improves with attention and CEL, peaking when $S = 8$ for both datasets, demonstrating the generalizability of our approach.

Model Variant	Attention	CEL	Subset Size	IQ-OTH/NCCD Dataset		HF Lung Cancer Dataset	
				Accuracy	F1 Score	Accuracy	F1 Score
CNN (Baseline)	✗	✗	–	0.953	0.947	0.890	0.883
+ Spatial Attention	✓	✗	–	0.962	0.959	0.901	0.894
+ CEL (w/o Attention)	✗	✓	8	0.967	0.966	0.906	0.901
+ CEL (S=2)	✓	✓	2	0.971	0.969	0.910	0.904
+ CEL (S=4)	✓	✓	4	0.980	0.978	0.917	0.911
+ CEL (S=6)	✓	✓	6	0.986	0.988	0.923	0.915
Full Model (S=8)	✓	✓	**8**	**0.992**	**0.995**	**0.930**	**0.920**

TABLE 10. Impact of margin parameter m on model performance. Results shown for both datasets using fixed $\lambda_{\text{CEL}} = 0.4$ and subset size $S = 8$.

Margin (m)	IQ-OTH/NCCD		HF Lung Cancer	
	Accuracy	F1 Score	Accuracy	F1 Score
0.3	0.973	0.975	0.912	0.908
0.5	0.984	0.986	0.921	0.912
0.6	0.989	0.991	0.926	0.917
0.7	**0.992**	**0.995**	**0.930**	**0.920**
0.8	0.990	0.992	0.927	0.918
0.9	0.983	0.985	0.921	0.913
1.1	0.975	0.977	0.914	0.906

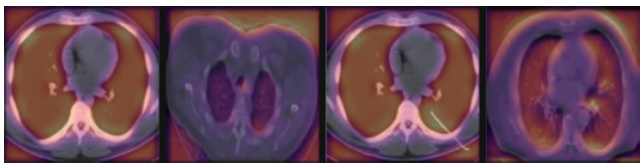


FIGURE 11. Analysis of misclassified examples from the HF Lung Cancer dataset. (a) Adenocarcinoma misclassified as Normal due to subtle, diffuse presentation; (b) Large Cell Carcinoma misclassified as Adenocarcinoma due to mixed morphological features; (c) Squamous Cell Carcinoma misclassified as Large Cell Carcinoma due to atypical location and appearance; (d) Normal case misclassified as Adenocarcinoma due to anatomical variations mistaken for nodules. Grad-CAM heatmaps reveal attention to diagnostically ambiguous regions.

typically present centrally, this case manifests peripherally with irregular borders. The model's attention focuses correctly on the lesion but misinterprets its classification due to the unusual location contradicting learned spatial priors. The final case (Figure 11d) is a Normal scan misclassified as Adenocarcinoma due to normal anatomical variations around the pulmonary vessels that mimic nodular appearance. Here, the model's attention incorrectly fixates on benign vascular structures, highlighting a limitation in distinguishing between normal anatomical variants and pathological findings.

These misclassifications reveal three primary error patterns: (1) detection failures for subtle, low-contrast abnormalities; (2) confusion between cancer subtypes with overlapping imaging characteristics; and (3) misinterpretation of normal anatomical variants. Notably, in all cases, the model's attention mechanism, to an extent, correctly identifies regions of interest, but either focuses on benign structures or fails to extract the specific features necessary for accurate classification. These findings suggest potential improvements through: (1) curriculum learning to progressively introduce subtle cases; (2) enhanced data augmentation targeting boundary cases between cancer subtypes; and (3) incorporation of multi-scale feature extraction to better distinguish fine-grained differences in tissue morphology. Additionally, integrating explicit anatomical priors could help prevent false positives from normal variants.

V. CONCLUSION

We introduced Contrastive Explainability Learning (CEL) as a novel approach to enhance interpretability in deep learning models for lung cancer detection. Unlike traditional methods that prioritize only accuracy, CEL integrates contrastive saliency loss, ensuring that model explanations remain structured and consistent across samples. Our lightweight CNN, trained with CEL, achieved strong performance on both tested datasets, recording 99.2% accuracy and 99.5% F1 score on the IQ-OTH/NCCD dataset and 93.0% accuracy on the more complex HF Lung Cancer benchmark. Importantly, CEL-guided saliency maps effectively highlight class-specific anatomical regions, providing improved transparency for clinical interpretation.

A. LIMITATIONS

Our study faces several important constraints. Both datasets lack comprehensive demographic information, preventing analysis of model performance across population subgroups - a critical concern given that medical AI systems can exhibit varying performance across age, sex, race, and socioeconomic factors. Additionally, our evaluation doesn't address variations in imaging protocols (radiation dose, slice thickness, scanner manufacturers) that commonly occur in clinical practice and may affect model generalizability. Our current approach also relies on 2D image slices rather than full volumetric data, potentially missing important 3D context for lung nodule characterization.

B. ADVERSARIAL ROBUSTNESS

CEL's structured explanation space introduces unique vulnerabilities to adversarial attacks that could preserve predictions while distorting saliency maps—particularly concerning in clinical settings where explanations inform medical decisions. While the contrastive constraints may provide inherent

defense by maintaining consistent attention patterns within classes, they could also create new attack surfaces through targeted perturbations designed to cross margin boundaries. An adversary could potentially craft perturbations that shift attention from true diagnostic regions to non-pathological areas, misleading clinicians even when classifications remain correct.

C. CLINICAL VALIDATION

While our results demonstrate that CEL produces consistent and discriminative explanations, we acknowledge the absence of formal validation by radiologists. Our preliminary discussions with clinical experts suggest that class-consistent attention patterns align with clinical reasoning, but systematic evaluation through blinded reviews remains essential. This validation should assess alignment between generated heatmaps and diagnostically relevant regions, potential improvements in diagnostic confidence, and comparative trust scores against other XAI methods.

D. FUTURE WORK

We plan to address these limitations through several directions: (1) evaluating on demographically diverse, multi-center cohorts with varied imaging protocols through a three-phase validation plan including retrospective multi-center evaluation, technical robustness assessment, and prospective clinical validation; (2) investigating adversarial robustness by developing explanation-specific evaluation metrics, defense mechanisms, and trust scores for both predictions and explanations; (3) conducting systematic clinical validation with radiologists through structured, blinded evaluations; (4) extending to 3D analysis through meaningful slice extraction and cross-slice consistency while maintaining computational feasibility; (5) exploring transferability to other cancer types and imaging modalities; (6) integrating multi-modal diagnostic pipelines combining CT, histopathological, and genomic data; and (7) optimizing for low-latency inference on edge devices to enable real-time screening in resource-constrained settings.

ACKNOWLEDGMENT

(V. Shravya and Meghana Sunil contributed equally to this work.)

REFERENCES

[1] M. Masud, N. Sikder, A.-A. Nahid, A. K. Bairagi, and M. A. AlZain, "A machine learning approach to diagnosing lung and colon cancer using a deep learning-based classification framework," *Sensors*, vol. 21, no. 3, p. 748, Jan. 2021.

[2] A. Rahman, P. Ghosh, A. D. Raha, S. Biswas, and A. K. Bairagi, "Focusing on subtleties: Class-specific attention-based deep learning for precise diagnosis of lung and colon cancers," in *Proc. 2nd Int. Conf. Mach. Intell. Emerg. Technol. (MIET)*, in Lecture Notes in Networks and Systems, M. Satu, M. S. Arefin, P. Lio', and M. Kaiser, Eds., Singapore: Springer, 2025, pp. 83–100.

[3] M. Fatan Serj, B. Lavi, G. Hoff, and D. Puig Valls, "A deep convolutional neural network for lung cancer diagnostic," 2018, *arXiv:1804.08170*.

[4] S. Mangal, A. Chaurasia, and A. Khajanchi, "Convolution neural networks for diagnosing colon and lung cancer histopathological images," 2020, *arXiv:2009.03878*.

[5] B. Ozdemir, E. Aslan, and I. Pacal, "Attention enhanced InceptionNeXt-based hybrid deep learning model for lung cancer detection," *IEEE Access*, vol. 13, pp. 27050–27069, 2025.

[6] I. Paçal and O. Attallah, "InceptionNeXt-transformer: A novel multi-scale deep feature learning architecture for multimodal breast cancer diagnosis," *Biomed. Signal Process. Control*, vol. 110, Jan. 2025, Art. no. 108116.

[7] I. Pacal and O. Attallah, "Hybrid deep learning model for automated colorectal cancer detection using local and global feature extraction," *Knowl.-Based Syst.*, vol. 319, Jun. 2025, Art. no. 113625.

[8] H. Cao, H. Liu, E. Song, G. Ma, X. Xu, R. Jin, T. Liu, and C.-C. Hung, "A two-stage convolutional neural networks for lung nodule detection," *IEEE J. Biomed. Health Informat.*, vol. 24, no. 7, pp. 2006–2015, Jul. 2020.

[9] S. K. B. Sangeetha, S. K. Mathivanan, P. Karthikeyan, H. Rajadurai, B. D. Shivahare, S. Mallik, and H. Qin, "An enhanced multimodal fusion deep learning neural network for lung cancer classification," *Syst. Soft Comput.*, vol. 6, Dec. 2024, Art. no. 200068.

[10] S. Shen, S. X. Han, D. R. Aberle, A. A. Bui, and W. Hsu, "An interpretable deep hierarchical semantic convolutional neural network for lung nodule malignancy classification," *Expert Syst. Appl.*, vol. 128, pp. 84–95, Aug. 2019.

[11] S. Tomassini, N. Falcionelli, P. Sernani, L. Burattini, and A. F. Dragoni, "Lung nodule diagnosis and cancer histology classification from computed tomography data by convolutional neural networks: A survey," *Comput. Biol. Med.*, vol. 146, Jul. 2022, Art. no. 105691.

[12] C. Hatt, C. Galban, W. Labaki, E. Kazerooni, D. Lynch, and M. Han, "Convolutional neural network based copd and emphysema classifications are predictive of lung cancer diagnosis," in *Proc. Int. Workshop Reconstruction Anal. Moving Body Organs*, Cham, Switzerland. Springer, 2018, pp. 302–309.

[13] W.-C. Wang, E. Ahn, D. Feng, and J. Kim, "A review of predictive and contrastive self-supervised learning for medical images," *Mach. Intell. Res.*, vol. 20, no. 4, pp. 483–513, Jun. 2023, doi: 10.1007/s11633-022-1406-4.

[14] D. Zeng, Y. Wu, X. Hu, X. Xu, H. Yuan, M. Huang, J. Zhuang, J. Hu, and Y. Shi, "Positional contrastive learning for volumetric medical image segmentation," in *Proc. MICCAI*, 2021, pp. 221–230.

[15] X. Hu, D. Zeng, X. Xu, and Y. Shi, "Semi-supervised contrastive learning for label-efficient medical image segmentation," in *Proc. MICCAI*, 2021, pp. 481–490.

[16] J. Xiang, Z. Li, W. Wang, Q. Xia, and S. Zhang, "Self-ensembling contrastive learning for semi-supervised medical image segmentation," 2021, *arXiv:2105.12924*.

[17] Y. Wu, D. Zeng, Z. Wang, Y. Shi, and J. Hu, "Federated contrastive learning for volumetric medical image segmentation," in *Proc. MICCAI*, 2021, pp. 367–377.

[18] Y. Lu, A. Jha, R. Deng, and Y. Huo, "Contrastive learning meets transfer learning: A case study in medical image analysis," *Proc. SPIE*, vol. 12033, pp. 729–736, Apr. 2022.

[19] L. Liu, A. I. Aviles-Rivero, and C.-B. Schönlieb, "Contrastive registration for unsupervised medical image segmentation," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 1, pp. 147–159, Jan. 2025.

[20] X. Hou, Z. Liu, X. Li, X. Li, S. Sang, and Y. Zhang, "MKCL: Medical knowledge with contrastive learning model for radiology report generation," *J. Biomed. Informat.*, vol. 146, Oct. 2023, Art. no. 104496.

[21] M. Toğaçar, B. Ergen, and Z. Cömert, "Detection of lung cancer on chest CT images using minimum redundancy maximum relevance feature selection method with convolutional neural networks," *Biocybernetics Biomed. Eng.*, vol. 40, no. 1, pp. 23–39, Jan. 2020.

[22] B. H. M. van der Velden, H. J. Kuijf, K. G. A. Gilhuijs, and M. A. Viergever, "Explainable artificial intelligence (XAI) in deep learning-based medical image analysis," *Med. Image Anal.*, vol. 79, Jul. 2022, Art. no. 102470.

[23] K. Borys, Y. A. Schmitt, M. Nauta, C. Seifert, N. Krämer, C. M. Friedrich, and F. Nensa, "Explainable AI in medical imaging: An overview for clinical practitioners – beyond saliency-based XAI approaches," *Eur. J. Radiol.*, vol. 162, May 2023, Art. no. 110786.

[24] K. Borys, Y. A. Schmitt, M. Nauta, C. Seifert, N. Krämer, C. M. Friedrich, and F. Nensa, "Explainable ai in medical imaging: An overview for clinical practitioners–saliency-based XAI approaches," *Eur. J. Radiol.*, vol. 162, Jan. 2023, Art. no. 110787.

[25] W. Jin, X. Li, M. Fatehi, and G. Hamarneh, "Generating post-hoc explanation from deep neural networks for multi-modal medical image analysis tasks," *MethodsX*, vol. 10, Jan. 2023, Art. no. 102009.
[26] J. Hou, S. Liu, Y. Bie, H. Wang, A. Tan, L. Luo, and H. Chen, "Self-eXplainable AI for medical image analysis: A survey and new outlooks," 2024, *arXiv:2410.02331*.
[27] D. Muhammad and M. Bendechache, "Unveiling the black box: A systematic review of explainable artificial intelligence in medical image analysis," *Comput. Struct. Biotechnol. J.*, vol. 24, pp. 542–560, Dec. 2024.
[28] S. S. Band, A. Yarahmadi, C.-C. Hsu, M. Biyari, M. Sookhak, R. Ameri, I. Dehzangi, A. T. Chronopoulos, and H.-W. Liang, "Application of explainable artificial intelligence in medical health: A systematic review of interpretability methods," *Informat. Med. Unlocked*, vol. 40, Jan. 2023, Art. no. 101286.
[29] A. Holzinger, C. Biemann, C. S. Pattichis, and D. B. Kell, "What do we need to build explainable AI systems for the medical domain?" 2017, *arXiv:1712.09923*.
[30] S. Mirzazadeh et al., "Atcon: Attention consistency regularization for robust visual explanations," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR) Workshops*, Jun. 2022.
[31] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2021, pp. 1–11.
[32] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 618–626.
[33] Y. Chang et al., "Your attention is all you need: Attention consistency for semi-supervised learning," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2020, pp. 244–260.
[34] F. Liu et al., "Injecting human knowledge into saliency-based explainable ai," in *Proc. Med. Image Comput. Comput.-Assist. Intervent. (MICCAI)*, 2022, pp. 345–355.
[35] A. Ghorbani et al., "Shield: Improving saliency-based interpretability by enforcing human-aligned explanations," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2019.
[36] S. Singla et al., "Explaining deep models through contrastive explanations," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2022.
[37] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.
[38] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo, "Swin transformer: Hierarchical vision transformer using shifted windows," 2021, *arXiv:2103.14030*.
[39] H. Touvron, M. Cord, M. Douze, F. Massa, A. Sablayrolles, and H. Jégou, "Training data-efficient image transformers & distillation through attention," 2020, *arXiv:2012.12877*.
[40] W. Xu, Y. Xu, T. Chang, and Z. Tu, "Co-scale conv-attentional image transformers," 2021, *arXiv:2104.06399*.
[41] C.-F. Chen, Q. Fan, and R. Panda, "CrossViT: Cross-attention multi-scale vision transformer for image classification," 2021, *arXiv:2103.14899*.
[42] B. Heo, S. Yun, D. Han, S. Chun, J. Choe, and S. J. Oh, "Rethinking spatial dimensions of vision transformers," 2021, *arXiv:2103.16302*.
[43] B. Graham, A. El-Nouby, H. Touvron, P. Stock, A. Joulin, H. Jégou, and M. Douze, "Levit: A vision transformer in convnet's clothing for faster inference," 2021, *arXiv:2104.01136*.
[44] X. Chu, Z. Tian, Y. Wang, B. Zhang, H. Ren, X. Wei, H. Xia, and C. Shen, "Twins: Revisiting the design of spatial attention in vision transformers," 2021, *arXiv:2104.13840*.
[45] S. Venkatraman, J. Singh Walia, and J. Dhanith P R, "SAG-ViT: A scale-aware, high-fidelity patching approach with graph attention for vision transformers," 2024, *arXiv:2411.09420*.
[46] H. Alyasriy and M. Al-Huseiny, "The IQ-OTH/NCCD lung cancer dataset," Tech. Rep., 2023.
[47] (2024). *Lung Cancer CT Scan Dataset*. [Online]. Available: https://huggingface.co/datasets/dorsar/lung-cancer,
[48] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," *Commun. ACM*, vol. 60, no. 6, pp. 84–90, May 2017, doi: 10.1145/3065386.
[49] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A ConvNet for the 2020s," 2022, *arXiv:2201.03545*.
[50] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 4700–4708.
[51] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," 2019, *arXiv:1905.11946*.
[52] C. Szegedy, W. Liu, Y. Jia, P. Sermanet, S. Reed, D. Anguelov, D. Erhan, V. Vanhoucke, and A. Rabinovich, "Going deeper with convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 1–9.
[53] M. Tan, B. Chen, R. Pang, V. Vasudevan, M. Sandler, A. Howard, and Q. V. Le, "MnasNet: Platform-aware neural architecture search for mobile," 2018, *arXiv:1807.11626*.
[54] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNetV2: Inverted residuals and linear bottlenecks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4510–4520.
[55] I. Radosavovic, R. Prateek Kosaraju, R. Girshick, K. He, and P. Dollár, "Designing network design spaces," 2020, *arXiv:2003.13678*.
[56] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," 2015, *arXiv:1512.03385*.
[57] N. Ma, X. Zhang, H.-T. Zheng, and J. Sun, "ShuffleNet v2: Practical guidelines for efficient CNN architecture design," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 122–138.
[58] F. N. Iandola, S. Han, M. W. Moskewicz, K. Ashraf, W. J. Dally, and K. Keutzer, "SqueezeNet: AlexNet-level accuracy with 50x fewer parameters and <0.5MB model size," 2016, *arXiv:1602.07360*.
[59] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.
[60] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 248–255.
[61] S. Woo, J. Park, J. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 3–19.
[62] P. W. Koh, T. Nguyen, Y. S. Tang, S. Mussmann, E. Pierson, B. Kim, and P. Liang, "Concept bottleneck models," in *Proc. 37th Int. Conf. Mach. Learn.*, Jul. 2020, pp. 5338–5348. [Online]. Available: https://proceedings.mlr.press/v119/koh20a.html
[63] B. Kim, M. Wattenberg, J. Gilmer, C. J. Cai, J. Wexler, F. Viégas, and R. Sayres, "Interpretability beyond feature attribution: Quantitative testing with concept activation vectors (TCAV)," in *Proc. 35th Int. Conf. Mach. Learn.*, Jul. 2017, pp. 2668–2677. [Online]. Available: https://proceedings.mlr.press/v80/kim18d.html
[64] M. T. Ribeiro, S. Singh, and C. Guestrin, "'Why should i trust you?': Explaining the predictions of any classifier," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, Aug. 2016, pp. 1135–1144, doi: 10.1145/2939672.2939778.
[65] M. Sundararajan, A. Taly, and Q. Yan, "Axiomatic attribution for deep networks," in *Proc. 34th Int. Conf. Mach. Learn.*, Aug. 2017, pp. 3319–3328. [Online]. Available: https://proceedings.mlr.press/v70/sundararajan17a.html
[66] A. Chattopadhay, A. Sarkar, P. Howlader, and V. N. Balasubramanian, "Grad-CAM++: Generalized gradient-based visual explanations for deep convolutional networks," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Mar. 2018, pp. 839–847.
[67] H. Wang, Z. Wang, M. Du, F. Yang, Z. Zhang, S. Ding, P. Mardziel, and X. Hu, "Score-CAM: Score-weighted visual explanations for convolutional neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2020, pp. 111–119.



V. SHRAVYA is currently pursuing the B.Tech. degree in computer science specializing in data science with VIT Chennai. She has published research papers in these areas, focusing on AI-driven solutions and data-driven decision-making. She aims to apply her knowledge to solve real-world challenges in AI and data science. Her research interests include artificial intelligence, machine learning, deep learning, and data science.



MEGHANA SUNIL is currently pursuing the B.Tech. degree in computer science specializing in data science with VIT Chennai. She has published research exploring intelligent systems and data-centric approaches. She is keen on using her skills to build impactful, real-world applications in the field of AI and data science. Her research interests include artificial intelligence, machine learning, deep learning, and big data analytics.



B. NATARAJAN received the Ph.D. degree from SASTRA University, Thanjavur, India. He is currently an Assistant Professor with Vellore Institute of Technology, Chennai. With a decade of academic teaching experience, he has authored numerous articles in reputed international journals. His research interests include computer vision, machine learning, deep learning, reinforcement learning, generative AI, explainable AI, quantum computing, and sign language development.



R. ELAKKIYA (Member, IEEE) received the bachelor's, master's, and Ph.D. degrees from Anna University. She securing a University First Rank and Gold Medal. She is an Assistant Professor with BITS Pilani, Dubai, and the Associate Head with APPCAIR. Recognized among the Top 2% World Scientists (Stanford-Elsevier, 2024). She's led 14 funded projects from DST-RFBR, DRDO, AICTE, Royal Society, UK, and industry partners. With more than 75 publications, six patents, and six books, she's also the Founder and the VP of Dubai ACM Chapter. An Intel OneAPI Innovator and Rising Star Educator, she's a frequent speaker and member of ACM and IAENG. Her research interests include AI, ML, and image processing.